

Hundred Days of Coding in C Programing

Day-10

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Step-1 Algorithm:

Convert Seconds into Days, Hours, Minutes and Seconds

- 1.Start.
- 2.Input the total time in seconds totalSeconds.
- 3.Calculate days: $\text{days} = \text{totalSeconds} / 86400$
- 4.Calculate the remaining seconds: $\text{remainingSeconds} = \text{totalSeconds} \% 86400$
- 5.Calculate hours: $\text{hours} = \text{remainingSeconds} / 3600$
- 6.Calculate the remaining seconds: $\text{remainingSeconds} = \text{remainingSeconds} \% 3600$
- 7.Calculate minutes: $\text{minutes} = \text{remainingSeconds} / 60$
- 8.Calculate seconds: $\text{seconds} = \text{remainingSeconds} \% 60$
- 9.Display days, hours, minutes and seconds.
- 10.Stop.

STEP 2 — Test Case Table

Test Case	Input (Seconds)	Expected Output
1	0	0 Days ,0 Hours, 0 Minutes , 0 Seconds
2	60	0 Days ,0 Hours, 1 Minutes , 0 Seconds
3	3600	0 Days ,1 Hours, 0 Minutes , 0 Seconds
4	86400	1 Days ,0 Hours, 0 Minutes , 0 Seconds
5	90061	1 Days ,1 Hours, 1 Minutes , 1 Seconds

STEP 3 — Pseudocode

BEGIN

INPUT totalSeconds

days $\leftarrow$ totalSeconds / 86400

remainingSeconds $\leftarrow$ totalSeconds % 86400

hours $\leftarrow$ remainingSeconds / 3600

remainingSeconds $\leftarrow$ remainingSeconds % 3600

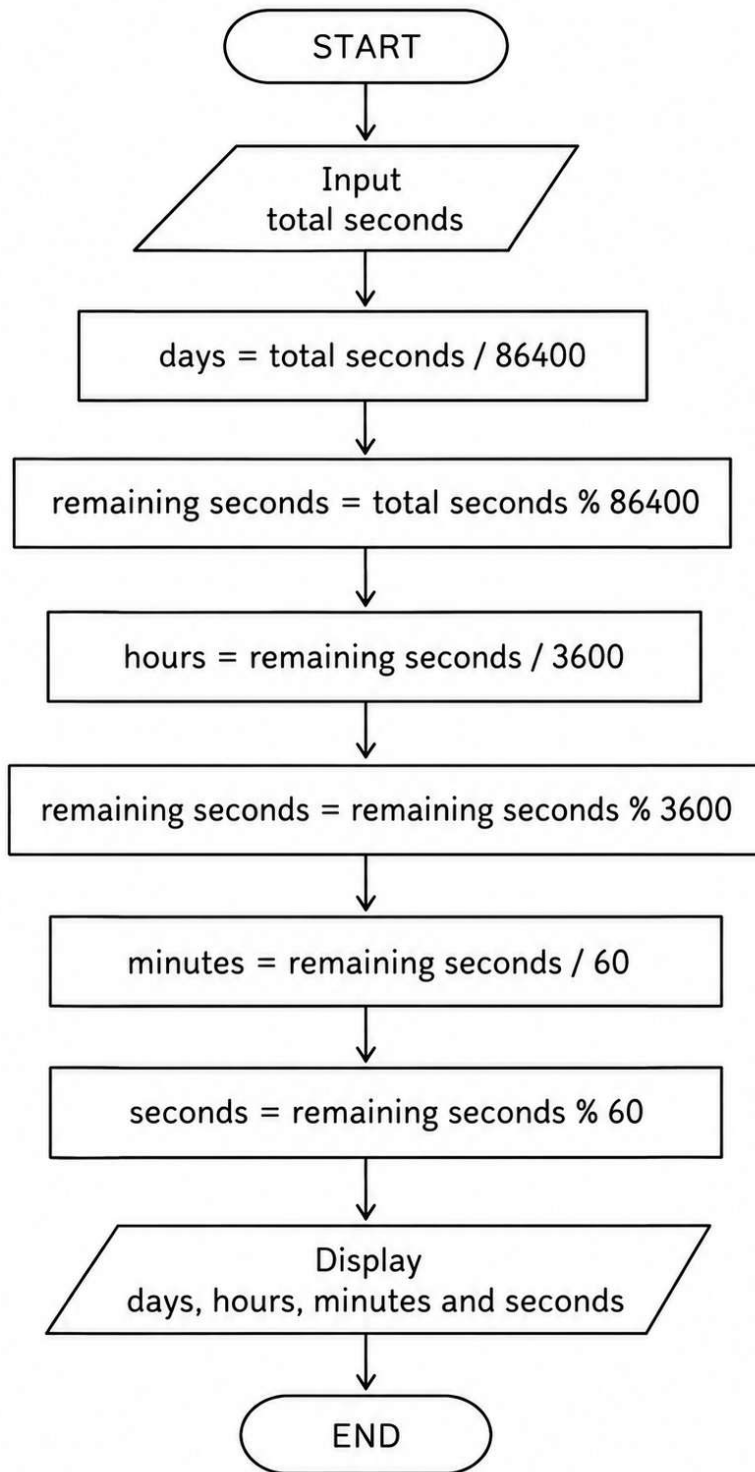
minutes $\leftarrow$ remainingSeconds / 60

seconds $\leftarrow$ remainingSeconds % 60

OUTPUT days, hours, minutes, seconds

END

STEP 4 — Flowchart



STEP 5 — C Program

```
#include <stdio.h>

int main()
{
    int totalSeconds, days, hours, minutes, seconds;
    int remainingSeconds;

    printf("Enter time in seconds: ");
    scanf("%d", &totalSeconds);

    days = totalSeconds / 86400;
    remainingSeconds = totalSeconds % 86400;

    hours = remainingSeconds / 3600;
    remainingSeconds = remainingSeconds % 3600;

    minutes = remainingSeconds / 60;
    seconds = remainingSeconds % 60;

    printf("Days = %d\n", days);
    printf("Hours = %d\n", hours);
    printf("Minutes = %d\n", minutes);
    printf("Seconds = %d\n", seconds);

    return 0;
}
```

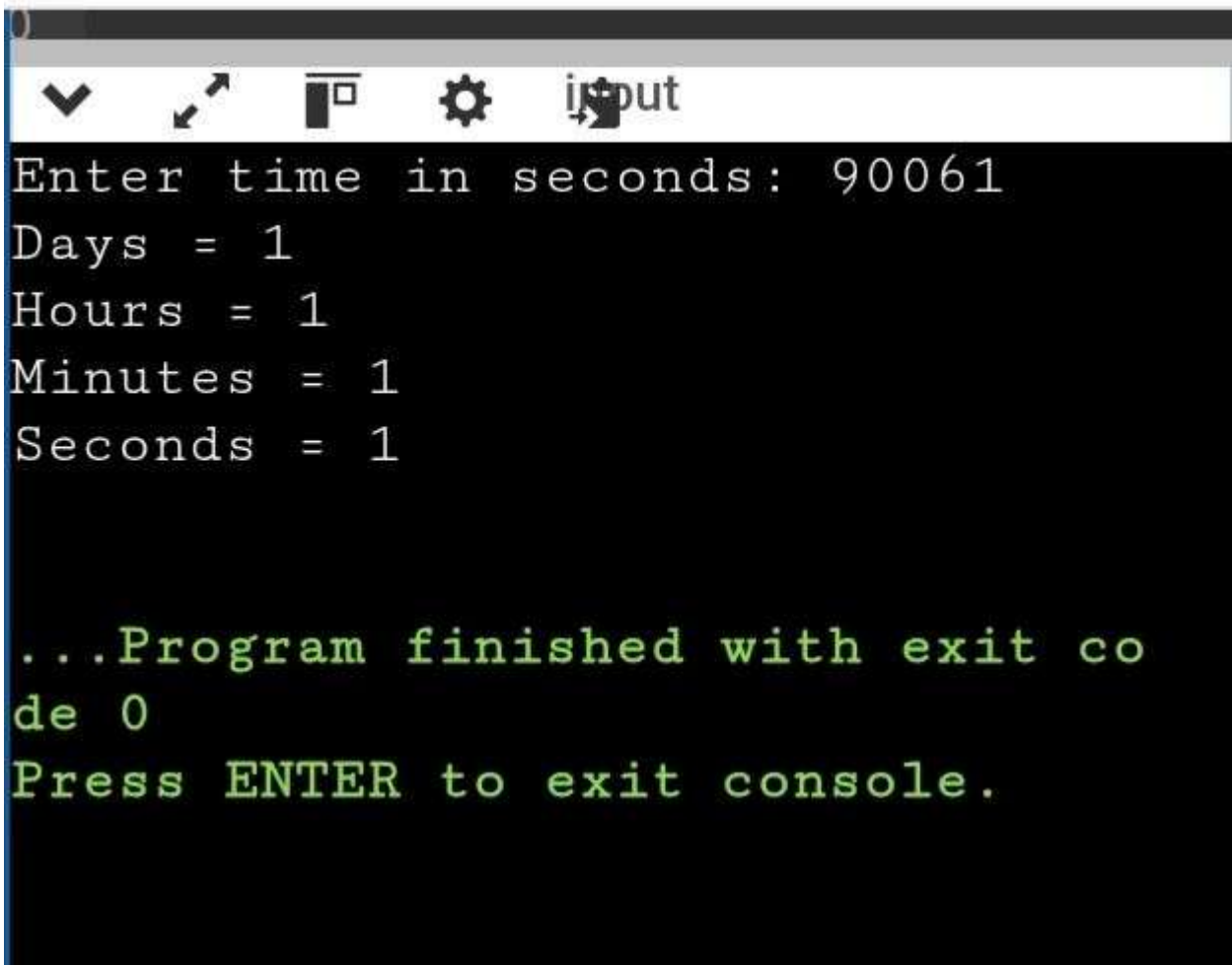
STEP 6 — Compile and Execute

Compiler Used: OnlineGDB

Programming Language: C

Program: Successfully compiled and executed.

Input: 90061 seconds

A screenshot of a terminal window with a dark background. The title bar at the top is light gray and contains several icons: a downward arrow, a cursor, a square, a gear, and the word 'input'. The terminal text is as follows:

```
Enter time in seconds: 90061
Days = 1
Hours = 1
Minutes = 1
Seconds = 1

...Program finished with exit co
de 0
Press ENTER to exit console.
```

Output: 1 Day, 1 Hour, 1 Minute, 1 Second Result:

The program was successfully compiled and executed. The output obtained is correct.

Step 7

Case 1:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int totalSeconds, days, hours, minutes, seconds;    int remainingSeconds;
6
7      printf("Enter time in seconds: ");    scanf("%d", &totalSeconds);
8
9      days = totalSeconds / 86400;    remainingSeconds = totalSeconds % 86400;
10
11     hours = remainingSeconds / 3600;    remainingSeconds = remainingSeconds % 3600;
12
13     minutes = remainingSeconds / 60;    seconds = remainingSeconds % 60;
14
15     printf("Days = %d\n", days);    printf("Hours = %d\n", hours);
16     printf("Minutes = %d\n", minutes);    printf("Seconds = %d\n", seconds);
17
18     return 0;
19 }
```

input

```
Enter time in seconds: 0
Days = 0
Hours = 0
Minutes = 0
Seconds = 0
```

Case 2:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int totalSeconds, days, hours, minutes, seconds;    int remainingSeconds;
6
7      printf("Enter time in seconds: ");    scanf("%d", &totalSeconds);
8
9      days = totalSeconds / 86400;    remainingSeconds = totalSeconds % 86400;
10
11     hours = remainingSeconds / 3600;    remainingSeconds = remainingSeconds % 3600;
12
13     minutes = remainingSeconds / 60;    seconds = remainingSeconds % 60;
14
15     printf("Days = %d\n", days);    printf("Hours = %d\n", hours);
16     printf("Minutes = %d\n", minutes);    printf("Seconds = %d\n", seconds);
17
18     return 0;
19 }
```

input

```
Enter time in seconds: 60
Days = 0
Hours = 0
Minutes = 1
Seconds = 0
```

Case 3:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int totalSeconds, days, hours, minutes, seconds;    int remainingSeconds;
6
7      printf("Enter time in seconds: ");    scanf("%d", &totalSeconds);
8
9      days = totalSeconds / 86400;    remainingSeconds = totalSeconds % 86400;
10
11     hours = remainingSeconds / 3600;    remainingSeconds = remainingSeconds % 3600;
12
13     minutes = remainingSeconds / 60;    seconds = remainingSeconds % 60;
14
15     printf("Days = %d\n", days);    printf("Hours = %d\n", hours);
16     printf("Minutes = %d\n", minutes);    printf("Seconds = %d\n", seconds);
17
18     return 0;
19 }
```

input

```
Enter time in seconds: 3600
Days = 0
Hours = 1
Minutes = 0
Seconds = 0
```


Case 4:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int totalSeconds, days, hours, minutes, seconds;    int remainingSeconds;
6
7      printf("Enter time in seconds: ");    scanf("%d", &totalSeconds);
8
9      days = totalSeconds / 86400;    remainingSeconds = totalSeconds % 86400;
10
11     hours = remainingSeconds / 3600;    remainingSeconds = remainingSeconds % 3600;
12
13     minutes = remainingSeconds / 60;    seconds = remainingSeconds % 60;
14
15     printf("Days = %d\n", days);    printf("Hours = %d\n", hours);
16     printf("Minutes = %d\n", minutes);    printf("Seconds = %d\n", seconds);
17
18     return 0;
19 }
```

input

```
Enter time in seconds: 86400
Days = 1
Hours = 0
Minutes = 0
Seconds = 0
```

Case 5:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int totalSeconds, days, hours, minutes, seconds;    int remainingSeconds;
6
7      printf("Enter time in seconds: ");    scanf("%d", &totalSeconds);
8
9      days = totalSeconds / 86400;    remainingSeconds = totalSeconds % 86400;
10
11     hours = remainingSeconds / 3600;    remainingSeconds = remainingSeconds % 3600;
12
13     minutes = remainingSeconds / 60;    seconds = remainingSeconds % 60;
14
15     printf("Days = %d\n", days);    printf("Hours = %d\n", hours);
16     printf("Minutes = %d\n", minutes);    printf("Seconds = %d\n", seconds);
17
18     return 0;
19 }
```

input

Enter time in seconds: 90061
Days = 1
Hours = 1
Minutes = 1
Seconds = 1

STEP 8 — Conclusion

Conclusion:

The C program successfully converts the given time in seconds into days, hours, minutes, and seconds. The program was compiled, executed, and tested using different test cases, and all the test cases produced the expected results.

Question 2

Write a C program to input a year and determine whether it is a leap year or not.

STEP 1 — Algorithm

- 1.Start.
- 2.Input the year.
- 3.Check whether the year is divisible by 400.
- 4.If yes, it is a leap year.
- 5.Otherwise, check whether the year is divisible by 100.
- 6.If yes, it is not a leap year.
- 7.Otherwise, check whether the year is divisible by 4.
- 8.If yes, it is a leap year.
- 9.Otherwise, it is not a leap year.
- 10.Display the result.
- 11.Stop.

STEP 2 — Test Case Table

Test Case	Input (Year)	Expected output
1	2024	Leap Year
2	2023	Not a leap year
3	2000	Leap Year
4	1900	Not a leap Year
5	2028	Leap Year

STEP 3 — Pseudocode

BEGIN

INPUT year

IF year % 400 == 0

 OUTPUT "Leap Year"

ELSE IF year % 100 == 0

 OUTPUT "Not a Leap Year"

ELSE IF year % 4 == 0

 OUTPUT "Leap Year"

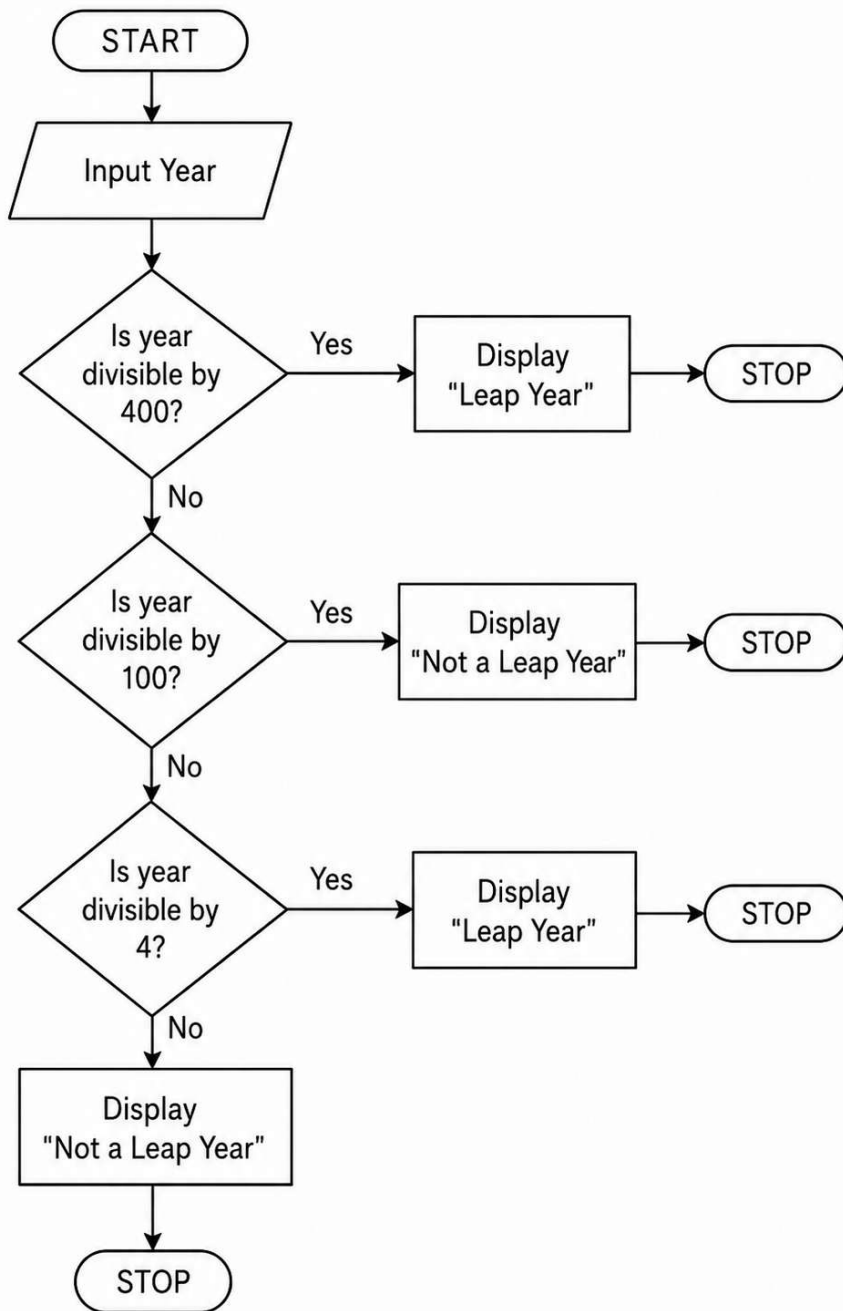
ELSE

 OUTPUT "Not a Leap Year"

END IF

END

STEP 4 — Flowchart



STEP 5 — C Program

```
#include <stdio.h>
```

```
int main()
```

```
{    int
```

```
year;
```

```
    printf("Enter a year: ");
```

```
    scanf("%d", &year);
```

```
    if (year % 400 == 0)
```

```
    {
```

```
        printf("Leap Year");
```

```
    }
```

```
    else if (year % 100 == 0)
```

```
    {
```

```
        printf("Not a Leap Year");
```

```
    }
```

```
    else if (year % 4 == 0)
```

```
    {
```

```
        printf("Leap Year");
```

```
    }
```

```
else
```

```
{
```

```
    printf("Not a Leap Year");
```

```
}
```

```
    return 0;
```

```
}
```

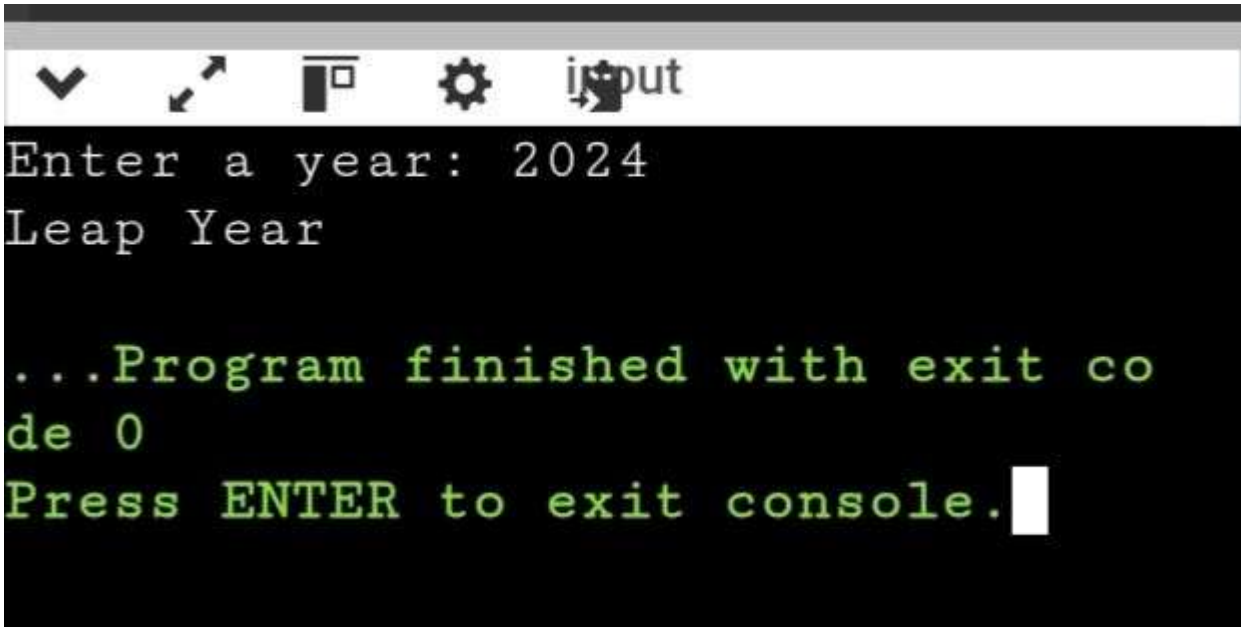
STEP 6 — Compile and Execute

Compiler Used: OnlineGDB

Programming Language: C

Input:

2024

A screenshot of a web-based IDE console. The top toolbar contains icons for a dropdown menu, a zoom-in icon, a zoom-out icon, a settings gear, and a cursor icon. The console text is as follows:
Enter a year: 2024
Leap Year

...Program finished with exit co
de 0
Press ENTER to exit console. [white cursor box]

Output:

Enter a year: 2024

Leap Year

Result:

The program was successfully compiled and executed in OnlineGDB, and the obtained output is correct.

STEP 7 — Test the Program

Case 1:

```
1  #include <stdio.h>
2
3  int main()
4  {      int year;
5
6          printf("Enter a year: ");      scanf("%d", &year);
7
8          if (year % 400 == 0)
9          {
10             printf("Leap Year");
11         }
12         else if (year % 100 == 0)
13         {
14             printf("Not a Leap Year");
15         }
16         else if (year % 4 == 0)
17         {
18             printf("Leap Year");
19         }
20         else
21         {
22             printf("Not a Leap Year");
23         }
24
25         return 0;
26     }
```

input

Enter a year: 2024
Leap Year

Case 2:

```
1  #include <stdio.h>
2
3  int main()
4  {   int year;
5
6      printf("Enter a year: ");   scanf("%d", &year);
7
8      if (year % 400 == 0)
9      {
10         printf("Leap Year");
11     }
12     else if (year % 100 == 0)
13     {
14         printf("Not a Leap Year");
15     }
16     else if (year % 4 == 0)
17     {
18         printf("Leap Year");
19     }
20     else
21     {
22         printf("Not a Leap Year");
23     }
24     return 0;
25 }
```

input

Enter a year: 2023
Not a Leap Year

Case 3:

```
1  #include <stdio.h>
2
3  int main()
4  {      int year;
5
6          printf("Enter a year: ");      scanf("%d", &year);
7
8          if (year % 400 == 0)
9          {
10             printf("Leap Year");
11         }
12         else if (year % 100 == 0)
13         {
14             printf("Not a Leap Year");
15         }
16         else if (year % 4 == 0)
17         {
18             printf("Leap Year");
19         }
20         else
21         {
22             printf("Not a Leap Year");
23         }
24         return 0;
25     }
```

input

Enter a year: 2000
Leap Year

Case 4:

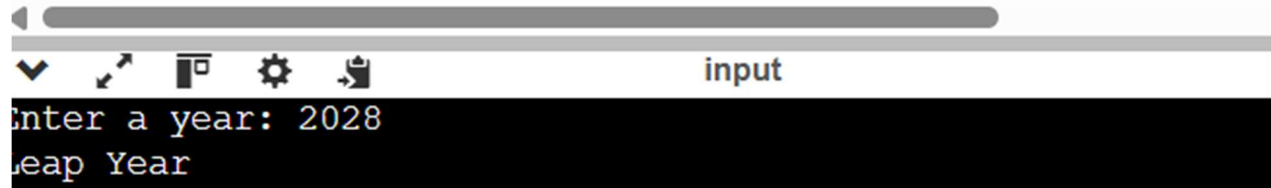
```
1  #include <stdio.h>
2
3  int main()
4  {
5      int year;
6
7      printf("Enter a year: ");    scanf("%d", &year);
8
9      if (year % 400 == 0)
10     {
11         printf("Leap Year");
12     }
13     else if (year % 100 == 0)
14     {
15         printf("Not a Leap Year");
16     }
17     else if (year % 4 == 0)
18     {
19         printf("Leap Year");
20     }
21     else
22     {
23         printf("Not a Leap Year");
24     }
25
26     return 0;
27 }
```

input

Enter a year: 1900
Not a Leap Year

Case 5:

```
1  #include <stdio.h>
2
3  int main()
4  {      int year;
5
6          printf("Enter a year: ");      scanf("%d", &year);
7
8          if (year % 400 == 0)
9          {
10             printf("Leap Year");
11         }
12         else if (year % 100 == 0)
13         {
14             printf("Not a Leap Year");
15         }
16         else if (year % 4 == 0)
17         {
18             printf("Leap Year");
19         }
20         else
21         {
22             printf("Not a Leap Year");
23         }
24
25         return 0;
26     }
```



The screenshot shows a terminal window with a dark background. The C code is displayed in a monospaced font with syntax highlighting. The code defines a function `main` that prompts the user to enter a year. It then uses a series of `if-else` statements to determine if the year is a leap year. The conditions are: `year % 400 == 0` (Leap Year), `year % 100 == 0` (Not a Leap Year), `year % 4 == 0` (Leap Year), and otherwise (Not a Leap Year). The program returns 0. Below the code, the terminal shows the prompt "Enter a year: 2028" and the output "Leap Year".

Step 8 Conclusion

The C program successfully determines whether a given year is a leap year or not. The program was compiled, executed, and tested with different test cases, and all test cases produced the expected results.